

Algebraic Derivation of the $SU(3)_c$ Gauge Structure

from the $\mathbb{Z}_3$ Triality of the $T^4/\mathbb{Z}_3$ Orbifold

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List of Symbols

Symbol	Meaning	Value / Unit
$\mathbb{Z}_3$	cyclic group of order 3	$\{e, g, g^2\}, g^3 = e$
ω	primitive $\mathbb{Z}_3$ phase factor	$e^{2\pi i/3}$
τ	generator of the $\mathbb{Z}_3$ action on T^4	unitary
P_k	projectors onto ω^k eigensectors	$k = 0, 1, 2$
$\mathfrak{su}(3)$	Lie algebra of $SU(3)$	dim 8
λ_a	Gell-Mann matrices, $a = 1, \dots, 8$	Hermitian, traceless
f_{abc}	$\mathfrak{su}(3)$ structure constants	totally antisymm.
T_R	triality representation index	$T_R \in \{0, 1, 2\} \bmod 3$
3, $\bar{3}$, 8	irreducible $SU(3)$ repr.	quark, antiquark, gluon
N_c	number of colour degrees of freedom	$N_c = 3$
α_s	strong coupling constant	$3\xi^{1/4}$ at m_τ (Doc. 160)
ξ	FFGFT geometry parameter	4/30000
[K]	derivable from ξ	status marker
[B]	algebraically proved	status marker
[S]	sketched / plausible	status marker

Summary

Doc. 321 carries out the algebraic derivation marked [S] open in Docs. 319, 320 and 322: the Gell-Mann matrices and the complete $SU(3)_c$ Lie algebra emerge from the $\mathbb{Z}_3$ triality of the $T^4/\mathbb{Z}_3$ orbifold. The proof proceeds in four steps: (1) the $\mathbb{Z}_3$ projectors on $L^2(T^4)$ yield a decomposition into three eigensectors; (2) the bilinear transitions between these sectors generate eight linearly independent operators; (3) these eight operators satisfy the $\mathfrak{su}(3)$ commutation relations; (4) the triality index distinguishes quarks ($T_R = 1, 2$) from gluons ($T_R = 0$) and explains confinement structurally.

Status markers

[K] derivable from ξ **[B]** algebraically proved **[S]** sketched/plausible **[SETZUNG]** axiomatic setting

1 Starting Point: $\mathbb{Z}_3$ as the Centre of $SU(3)$

The centre of a Lie group

The centre $Z(G)$ of a group G consists of all elements that commute with every other element: $Z(G) = \{z \in G : zg = gz \forall g \in G\}$.

[K] For $SU(3)$:

$$Z(SU(3)) = \mathbb{Z}_3 = \{\mathbf{1}, \omega\mathbf{1}, \omega^2\mathbf{1}\}, \quad \omega = e^{2\pi i/3}. \quad (1)$$

[K] This is not a coincidence: $\mathbb{Z}_3 = Z(SU(3))$ means that any $\mathbb{Z}_3$ symmetry embeddable into the unitary group leads automatically into the $SU(3)$ structure. The $\mathbb{Z}_3$ action on $T^4/\mathbb{Z}_3$ is therefore not an additional FFGFT assumption but the minimal unitary extension of the $\mathbb{Z}_3$ centre to a full gauge group.

Triality: the $\mathbb{Z}_3$ charge of $SU(3)$ representations

[B] Every irreducible representation of $SU(3)$ has a *triality* $T_R \in \{0, 1, 2\}$, defined by the $\mathbb{Z}_3 = Z(SU(3))$ action:

$$\omega \mathbf{1} \xrightarrow{\text{repr. } R} \omega^{T_R} \cdot \mathbf{1}_R. \quad (2)$$

Representation	Dimension	Triality T_R	Physical role
1	1	0	singlet (colour-neutral)
8	8	0	gluons (adjoint)
3	3	1	quarks
$\bar{3}$	3	2	antiquarks

[K] Confinement is equivalent to requiring all observable states to have triality $T_R = 0$ (Doc. 049: "nodes cannot exist alone, must form neutral combinations").

.2 The $\mathbb{Z}_3$ Sector Decomposition on $L^2(T^4)$

Three projectors from one $\mathbb{Z}_3$ action

[K] The $\mathbb{Z}_3$ group $\{e, g, g^2\}$ acts on T^4 through the unitary operator τ ($\tau^3 = \text{id}$). On $L^2(T^4)$ this induces three orthogonal projectors:

$$P_k = \frac{1}{3} \sum_{j=0}^2 \omega^{-jk} \tau^j, \quad k = 0, 1, 2. \quad (3)$$

[B] These projectors are pairwise orthogonal and complete:

$$P_k^2 = P_k, \quad P_k P_l = \delta_{kl} P_k, \quad P_0 + P_1 + P_2 = \text{id}. \quad (4)$$

Proof of idempotency: $P_k^2 = \frac{1}{9} \sum_{j,j'} \omega^{-k(j+j')} \tau^{j+j'} = \frac{1}{9} \sum_n 3 \omega^{-kn} \tau^n = P_k. \checkmark$

The three eigensectors and their modes

[K] In the Fourier basis $e^{in \cdot x/R}$ the $\mathbb{Z}_3$ action $(x_1, x_2, x_3, x_4) \mapsto (x_2, x_3, x_1, x_4)$ maps (n_1, n_2, n_3) to (n_3, n_1, n_2) . The sum $n_1 + n_2 + n_3$ is unchanged under τ and therefore does not fix the sector. A single Fourier mode is an eigenvector of τ only for $n_1 = n_2 = n_3$ (then in $\mathcal{H}_0$). The eigenstates are orbit combinations

$$\psi_{n,k} = \sum_{j=0}^2 \omega^{-jk} \tau^j e^{in \cdot x/R} \in \mathcal{H}_k, \quad \tau \psi_{n,k} = \omega^k \psi_{n,k}; \quad (5)$$

each three-element orbit $\{n, \tau n, \tau^2 n\}$ yields one state each in $\mathcal{H}_0, \mathcal{H}_1$ and $\mathcal{H}_2$. Thus the elements of the orbit of $(1, 0, 0)$ all have sum 1, and the orbit still yields one state in each of the three sectors:

- $\mathcal{H}_0$: colour-neutral (gluons, singlets).
- $\mathcal{H}_1$: quarks.
- $\mathcal{H}_2$: antiquarks.

.3 Algebraic Construction of the Eight Gell-Mann Operators

Bilinear transitions between sectors

[B] The diagonal basis (e_0, e_1, e_2) with $\tau e_k = \omega^k e_k$ corresponds to the three colour states (Red = e_0 , Green = e_1 , Blue = e_2). The $3^2 - 1 = 8$ traceless Hermitian operators on $\mathbb{C}^3$ are constructed as:

$$\begin{aligned} \text{Transitions } \mathcal{H}_0 \leftrightarrow \mathcal{H}_1: \quad T_1 &= \frac{1}{2}(|e_0\rangle\langle e_1| + |e_1\rangle\langle e_0|), & T_2 &= \frac{i}{2}(|e_1\rangle\langle e_0| - |e_0\rangle\langle e_1|). \\ \text{Transitions } \mathcal{H}_0 \leftrightarrow \mathcal{H}_2: \quad T_4 &= \frac{1}{2}(|e_0\rangle\langle e_2| + |e_2\rangle\langle e_0|), & T_5 &= \frac{i}{2}(|e_2\rangle\langle e_0| - |e_0\rangle\langle e_2|). \\ \text{Transitions } \mathcal{H}_1 \leftrightarrow \mathcal{H}_2: \quad T_6 &= \frac{1}{2}(|e_1\rangle\langle e_2| + |e_2\rangle\langle e_1|), & T_7 &= \frac{i}{2}(|e_2\rangle\langle e_1| - |e_1\rangle\langle e_2|). \\ \text{Diagonal operators:} \quad T_3 &= \frac{1}{2}(|e_0\rangle\langle e_0| - |e_1\rangle\langle e_1|), & T_8 &= \frac{1}{2\sqrt{3}}(|e_0\rangle\langle e_0| + |e_1\rangle\langle e_1| - 2|e_2\rangle\langle e_2|). \end{aligned}$$

Identification with the Gell-Mann matrices

[B] $T_a = \frac{1}{2}\lambda_a$ with:

$$\begin{aligned} \lambda_1 &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \lambda_2 &= \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \lambda_3 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ \lambda_4 &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, & \lambda_5 &= \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, & \lambda_6 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \\ \lambda_7 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, & \lambda_8 &= \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \end{aligned} \tag{6}$$

.4 Proof of the $\mathfrak{su}(3)$ Commutation Relations

Normalisation condition

[B] Probe for $a = b = 1$:

$$\text{tr}(T_1 T_1) = \frac{1}{4} \text{tr}(\lambda_1^2) = \frac{1}{4} \text{tr} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \frac{2}{4} = \frac{1}{2}. \quad \checkmark \tag{7}$$

Selected commutators

[B]

$$[T_1, T_2] = iT_3:$$

$$[T_1, T_2] = \frac{1}{4}[\lambda_1, \lambda_2] = \frac{i}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = iT_3. \quad \checkmark \tag{8}$$

$[T_1, T_4] = \frac{i}{2}T_7$: T_1 connects $\mathcal{H}_0 \leftrightarrow \mathcal{H}_1$, T_4 connects $\mathcal{H}_0 \leftrightarrow \mathcal{H}_2$, so their commutator falls in the $\mathcal{H}_1 \leftrightarrow \mathcal{H}_2$ sector. The imaginary transition T_7 appears (not T_6):

$$[T_1, T_4] = \frac{1}{4}[\lambda_1, \lambda_4] = \frac{i}{2}T_7. \quad \checkmark \tag{9}$$

Structure constant: $f_{147} = 1/2$ (canonical $\mathfrak{su}(3)$ value).

$[T_3, T_8] = 0$: both diagonal; diagonal matrices commute. ✓

T_3 and T_8 generate the Cartan subalgebra $\mathfrak{h} \subset \mathfrak{su}(3)$ (maximum abelian subalgebra, dimension 2).

Main result Doc. 321 [B]

The eight bilinear transition operators between the three $\mathbb{Z}_3$ eigensectors of $L^2(T^4)$ satisfy the $\mathfrak{su}(3)$ Lie algebra relations $[T_a, T_b] = if_{abc}T_c$ with the canonical structure constants and the normalisation condition $\text{tr}(T_a T_b) = \frac{1}{2}\delta_{ab}$. The $SU(3)_c$ gauge group emerges algebraically necessarily from the $\mathbb{Z}_3$ triality of the $T^4/\mathbb{Z}_3$ orbifold.

.5 Physical Interpretation

Eight gluons as transition operators

[K] The eight gluons correspond to the eight bilinear transition operators. Gluons have triality $T_R = 0$ (adjoint representation): the centre element $\omega \mathbf{1}$ acts trivially on the adjoint. In the sector decomposition only the diagonal operators T_3, T_8 have $\Delta k = 0$; the off-diagonal generators $T_1, T_2, T_4, \dots, T_7$ shift k by $\pm 1 \pmod{3}$ (e.g. T_1 connects $\mathcal{H}_0$ and $\mathcal{H}_1$). Triality 0 of the gluons is a property of the representation and does not follow from $\Delta k = 0$ in this decomposition. A quark (triality 1) emitting a gluon (triality 0) remains a quark (triality 1).

Confinement as triality selection

[K]

State	Composition	Triality	Observable?
Meson ($q\bar{q}$)	$T_R = 1 + 2$	$0 \pmod{3}$	Yes
Baryon (qqq)	$T_R = 1 + 1 + 1$	$0 \pmod{3}$	Yes
Single quark	$T_R = 1$	1	No

[B] A single quark has $T_R = 1 \neq 0$ and is therefore not a $\mathbb{Z}_3$ singlet. Its state function is not localised in $\mathcal{H}_0$ and cannot propagate as a free asymptotic state – the geometric confinement criterion (Doc. 049 in algebraic form).

.6 Connection to the Strong Coupling α_s

[K] Doc. 160 derives the strong coupling from orbifold geometry. With $N_c = 3$ now algebraically proved (three $\mathbb{Z}_3$ eigensectors):

$$\alpha_s(m_\tau) = N_c \cdot \xi^{1/(N_c+1)} = 3 \cdot \xi^{1/4} = 3 \cdot \left(\frac{4}{30000}\right)^{1/4} \approx 0.322. \quad (10)$$

PDG: $\alpha_s(m_\tau) \approx 0.330$. Deviation: -2.3% .

[B] The factor $N_c = 3$ in Eq. (10) was an input in Doc. 160. Doc. 321 derives it algebraically: N_c is the order of the $\mathbb{Z}_3$ group, i.e. the number of eigensectors of the $\mathbb{Z}_3$ action on $L^2(T^4)$. Eq. (10) is now fully parameter-free.

[K] Why $N_c = 3$ and not $N_c = 2$ or $N_c = 4$:

- $N_c = 3$ is the order of $\mathbb{Z}_3 = Z(SU(3))$ – the centre of the smallest simple compact non-abelian Lie group whose centre contains $\mathbb{Z}_3$; E_6 (centre $\mathbb{Z}_3$) and $SU(3k)$ (centre $\mathbb{Z}_{3k}$) also contain $\mathbb{Z}_3$ in their centre (the order of the centre is the determinant of the Cartan matrix: $A_2: 3$, $A_5: 6$, $A_8: 9$, $E_6: 3$).
- D_4 triality (Doc. 314) enforces exactly three classes of spin representations.
- For $\mathbb{Z}_n$ with $n > 3$ the $T^4/\mathbb{Z}_n$ orbifold has fewer independent fixed points, destroying the neutrino sector (Doc. 320).

.7 $SU(2)_L$ and $U(1)_Y$ from the Orbifold Sector Structure

Remaining torus degrees of freedom after $\mathbb{Z}_3$ projection

[K] The $\mathbb{Z}_3$ projection acts on the first three torus directions (x_1, x_2, x_3) . The fourth direction x_4 is $\mathbb{Z}_3$ -invariant and remains unfolded:

$$T^4/\mathbb{Z}_3 = \underbrace{T^3/\mathbb{Z}_3}_{SU(3)_c} \times \underbrace{T^1}_{U(1)_Y}. \quad (11)$$

Within the $\mathbb{Z}_3$ -invariant sector $\mathcal{H}_0$ a $\mathbb{Z}_2$ symmetry between the ω - and ω^2 -sectors ($\mathcal{H}_1$ and $\mathcal{H}_2$) provides the source of $SU(2)_L$.

$U(1)_Y$ from the fourth torus direction

[B] The fourth torus direction carries a global $U(1)$ phase symmetry $x_4 \mapsto x_4 + \alpha$ with generator $\hat{Y} = \partial/\partial\alpha|_{\alpha=0}$. It acts as $\hat{Y}|n_4\rangle = n_4|n_4\rangle$, quantising hypercharge by the fourth winding number. Compatibility with $\mathbb{Z}_3$ forces $Y \in \frac{1}{3}\mathbb{Z}$.

$SU(2)_L$ from the $\mathbb{Z}_2$ pairing of sectors

[B] Complex conjugation $\mathcal{C} : \mathcal{H}_1 \leftrightarrow \mathcal{H}_2$ is a $\mathbb{Z}_2$ symmetry. In the reduced 2×2 sub-algebra of the $\mathcal{H}_1$ - $\mathcal{H}_2$ transitions it generates the Pauli matrices $\sigma_1, \sigma_2, \sigma_3$ with $[\sigma_i/2, \sigma_j/2] = i\epsilon_{ijk}\sigma_k/2$ – the $\mathfrak{su}(2)$ algebra, yielding $SU(2)_L$.

Weinberg angle from the trace formula of the five orbifold states

[B] The five FFGFT orbifold states and their quantum numbers:

State	Sector	T_3	Q	$Y = 2(Q - T_3)$
d_R, d_G, d_B	$\mathcal{H}_1$ (triality 1)	0	-1/3	-2/3
e^-	$\mathcal{H}_2$ (triality 2)	-1/2	-1	-1
ν_e	$\mathcal{H}_0$ (triality 0)	+1/2	0	-1

The Weinberg angle follows from the trace formula in the 5-plet:

$$\sin^2 \theta_W = \frac{\text{tr}[T_3^2]}{\text{tr}[Q^2]} = \frac{3 \cdot 0 + (1/2)^2 + (1/2)^2}{3 \cdot (1/3)^2 + 1^2 + 0^2} = \frac{1/2}{4/3} = \boxed{\frac{3}{8} = 0.375}. \quad (12)$$

[B] This is a direct algebraic consequence of the orbifold sector structure with $N_c = 3$ and coincides with the classical $SU(5)$ -GUT result of Georgi and Glashow (1974), now derived from FFGFT geometry.

RG running to M_Z

[K] The measured value $\sin^2 \theta_W(M_Z) = 0.2312$ (PDG) differs from the GUT-point value by $\Delta \sin^2 \theta_W = 3/8 - 0.2312 = 0.1438$, which is the well-known RG running from M_{GUT} to M_Z .

[K] The formal derivation of the dependence of $\Delta \sin^2 \theta_W$ on ξ is carried out in Doc. 323. The result $\sin^2 \theta_W(M_Z) = 3/8 - \frac{55\alpha_{\text{em}}}{24\pi} [\ln(m_{\text{Pl}}/M_Z) + \frac{19}{12} \ln \xi] = 0.2308$ deviates -0.19% from PDG.

Status balance $SU(2)_L \times U(1)_Y$

$U(1)_Y$: **[B]** from the fourth torus direction.

$SU(2)_L$: **[B]** from the $\mathbb{Z}_2$ pairing $\mathcal{H}_1 \leftrightarrow \mathcal{H}_2$.

$\sin^2 \theta_W|_{\text{GUT}} = 3/8$: **[B]** from the trace formula of the five orbifold states.

$\sin^2 \theta_W(M_Z) = 0.2308$: **[K]** RG running derived from ξ in Doc. 323 (-0.19%).

Complete gauge group picture

Factor	Origin	Generators	Status
$SU(3)_c$	$\mathbb{Z}_3$ triality on $T^3 \subset T^4$	8	[B] §3–4
$SU(2)_L$	$\mathbb{Z}_2$ pairing $\mathcal{H}_1 \leftrightarrow \mathcal{H}_2$	3	[B] §7.3
$U(1)_Y$	4th torus direction T^1	1	[B] §7.2
$\sin^2 \theta_W _{\text{GUT}} = 3/8$	trace formula	—	[B] §7.4
$\sin^2 \theta_W(M_Z)$	RG running	—	[K] Doc. 323
Full SM gauge group	$SU(3)_c \times SU(2)_L \times U(1)_Y$	12	[B]/[K]

.8 Cross-References and Corpus

Table 1: Derivation chain Doc. 321 and corpus references

Statement	Doc.	Status
$\mathbb{Z}_3 = Z(SU(3))$ (centre)	group theory	[B]
Triality as $\mathbb{Z}_3$ charge	repr. theory	[B]
$\mathbb{Z}_3$ projectors on $L^2(T^4)$	this Doc.	[B]
Three eigensectors $\mathcal{H}_0, \mathcal{H}_1, \mathcal{H}_2$	this Doc.	[K]
Eight bilinear transition operators	this Doc.	[B]
$\mathfrak{su}(3)$ commutation relations	this Doc.	[B]
Gell-Mann matrices from sector transitions	this Doc.	[B]
$N_c = 3$ algebraically derived	this Doc.	[B]
$\alpha_s(m_\tau) = 3\xi^{1/4}$ (now complete)	Doc. 160 + this	[K]
Confinement as triality selection	Doc. 049, this	[K]
D_4 triality, $\mathbb{Z}_3$ centre	Doc. 314	[K]
Gluon topology (flux tubes)	Docs. 145, 319	[S]
$U(1)_Y$ from 4th torus direction	this Doc.	[B]
$SU(2)_L$ from $\mathbb{Z}_2$ sector pairing	this Doc.	[B]
$\sin^2 \theta_W _{\text{GUT}} = 3/8$ (trace formula)	this Doc.	[B]
$\sin^2 \theta_W(M_Z)$ RG running from ξ	Doc. 323	[K]

Overall status Doc. 321

The complete algebraic derivation of the $SU(3)_c$ gauge structure is complete **[B]**. $N_c = 3$ is algebraically necessary; $\alpha_s(m_\tau) = 3\xi^{1/4}$ is fully parameter-free. $U(1)_Y$ (fourth torus direction) and $SU(2)_L$ ($\mathbb{Z}_2$ sector pairing) are algebraically derived **[B]**. The Weinberg angle at the GUT point $\sin^2 \theta_W = 3/8$ follows from the trace formula of the five orbifold states **[B]**. The full SM gauge group $SU(3)_c \times SU(2)_L \times U(1)_Y$ is algebraically founded. Doc. 323 derives the RG running $\sin^2 \theta_W(\text{GUT}) \rightarrow \sin^2 \theta_W(M_Z)$ from ξ **[K]**, deviation -0.19% .